

BIKKINA ABHIRAM CHOUDHARY

Embedded Systems & Firmware Engineer

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PROFESSIONAL SUMMARY

B.Tech ECE student specializing in embedded firmware development and IoT systems, with proven expertise in bare-metal C/C++ programming for ARM Cortex-M microcontrollers (STM32, ESP32, LPC2148) and hardware communication protocols (UART, SPI, I2C, GPIO, PWM). Delivered end-to-end embedded projects including an IoT-based cardiovascular disease detection system and an ESP32-CAM surveillance platform, translating hardware specifications into optimized, interrupt-driven firmware. Holds NI CLAD certification with internship experience in ARM-based firmware and HAL/LL driver development using Keil µVision. Actively advancing in FreeRTOS, BSP development, and Embedded Linux. Seeking a full-time Embedded Software Engineer or Firmware Engineer role to contribute to reliable, low-level hardware software development.

TECHNICAL SKILLS

Languages: C (Expert), Embedded C (Expert), C++ (Proficient), Python (Intermediate), Assembly (ARM, basic)

Microcontrollers: STM32 (Cortex-M3/M4), ESP32, LPC2148, Arduino Uno, 8051, ARM Cortex-M

Firmware Dev: Bare-metal firmware, HAL/LL drivers, BSP development, Register-level programming, Memory-mapped I/O, Bootloaders (basic)

Protocols: UART, SPI, I2C, GPIO, PWM, ADC/DAC, Bluetooth (HC-05), Wi-Fi, MQTT, CAN (learning)

RTOS / OS: FreeRTOS (learning), Task scheduling, Semaphores, Embedded Linux (Yocto/Buildroot basics)

Tools & IDEs: Keil µVision, STM32CubeIDE, Arduino IDE, STM Studio, EasyEDA, Tinkercad

Debug & Test: GDB/OpenOCD, JTAG/SWD, Logic Analyzer, Oscilloscope

Cloud / IoT: Blynk, ThingSpeak, MQTT,

Other: Git, LabVIEW (CLAD Certified), EasyEDA PCB Design

PROJECTS

IoT-Based Cardiovascular Disease Detection System

Jan 2025 – Feb 2025

ESP32 · MAX30102 · AD8232 ECG · SPI · I2C · MQTT · ThingSpeak · Blynk · Wi-Fi

- Engineered a non-invasive cardiovascular disease (CVD) early-detection system using ESP32, fusing data from MAX30102 (PPG-based SpO2/heart rate) and AD8232 ECG module via I2C and SPI to extract clinical cardiac biomarkers in real time.
- Implemented Heart Rate Variability (HRV) analysis and R-peak detection algorithm in Embedded C to identify arrhythmia patterns (bradycardia, tachycardia) with ~88% detection accuracy validated against reference ECG traces.
- Designed a multi-threshold alert engine in firmware that classifies cardiac risk into three severity levels (Normal / Warning / Critical) and triggers instant Blynk push notifications, reducing alert response time to under 3 seconds over Wi-Fi.
- Built an MQTT pipeline to stream time-stamped cardiac data to ThingSpeak cloud at 50 Hz, enabling remote cardiologist review and enabling longitudinal trend analysis across 7-day rolling datasets.
- Optimized firmware with interrupt-driven ADC sampling and digital band-pass filtering (0.5–40 Hz), reducing signal noise by 45% and cutting false-positive CVD alerts by 30% compared to raw unfiltered readings.

ESP32-CAM Wireless Surveillance System

Oct 2024 – Dec 2024

ESP32-CAM · OV2640 · Wi-Fi · HTTP Server · Motion Detection · Embedded C

- Engineered a low-cost wireless surveillance camera using ESP32-CAM and OV2640 image sensor, streaming HD video (1600×1200) over Wi-Fi via an onboard HTTP server.
- Implemented frame-differencing-based motion detection algorithm in firmware, reducing false alerts by 60% and triggering push notifications only on actual motion events.
- Configured camera frame buffer with DMA transfers for real-time JPEG encoding, achieving stable 15–20 fps streaming with under 80 ms end-to-end latency on a local network.
- Improved overall security monitoring efficiency by 30% compared to the baseline passive camera system through intelligent event-triggered capture and logging.

Arduino-Based Home Automation System (Android + Bluetooth)

Mar 2023 – Apr 2023

Arduino Uno · HC-05 Bluetooth · Relay Module · MIT App Inventor · UART

- Developed a smart home automation system enabling control of 4 AC appliances via an Android app, communicating over UART/Bluetooth (HC-05) at 9600 baud.
- Wrote Arduino firmware to parse AT commands from the HC-05 module and toggle relay outputs with <50 ms response time, ensuring responsive and reliable switching.
- Designed the Android control interface using MIT App Inventor with on/off toggles and real-time device status feedback, reducing manual switch interaction by ~80% in testing scenarios.

INTERNSHIP EXPERIENCE

Embedded Systems Engineer | InternzLearn

May 2024 – Sep 2024

- Developed embedded C firmware for ARM Cortex-M3/M4 microcontrollers, implementing GPIO control, timer PWM, UART communication, and interrupt service routines from scratch.
- Simulated and validated hardware designs virtually before flashing to physical targets using Keil µVision, reducing hardware debug cycles by ~30%.
- Performed code optimization (loop unrolling, register-level access, stack analysis) to reduce firmware footprint by 15% and meet strict memory constraints on resource-limited MCUs.
- Gained hands-on experience in system-level integration for embedded IoT applications, including sensor interfacing, data parsing, and real-time control loops.

EDUCATION

B.Tech — Electronics & Communication Engineering | KL University (KLEF), Guntur

Aug 2023 – Present

CGPA: 8.3 / 10.0 | Relevant coursework: Microprocessors & Microcontrollers, VLSI Design, Embedded Systems, Digital Signal Processing, IoT

Diploma — Electronics & Communication Engineering | Vikash Polytechnic College, Bargarh

Jun 2020 – Oct 2023

Percentage: 72.5%

Secondary School Certificate (Class X) — CBSE | The Kosala School, Nua Khairpali

Jun 2019 – May 2020

Percentage: 56%

CERTIFICATIONS

- **NI Certified LabVIEW Associate Developer (CLAD)** — National Instruments, Feb 2024
- **IIoT Programming & Automation** — Tessolve Semiconductors, April 2025